

Error Analysis Report

Project: AI-Powered Spectrum Anomaly Detection

Model: EfficientNet-B0

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1. Test Results Summary

- Total Test Samples: 1,179
- Overall Accuracy: 93.47%
- Total Errors: 77 (6.53%)

2. Errors per Class

- Drone: 64 errors (misclassified as Jamming)
- Normal: 0 errors
- Jamming: 0 errors

3. Error Analysis

The model achieves perfect classification for Normal and Jamming signals (100% recall). The only errors come from the Drone class, where 64 out of 393 Drone samples (16.3%) were misclassified as Jamming. This is primarily due to the multi-modal nature of drone signals. Different drone types have varying spectral signatures, and some overlap with jamming patterns.

Main causes for Drone misclassification:

- * Drone signals are not a single cluster in feature space
- * Some drone types share characteristics with jamming signals
- * Limited diversity of drone data compared to normal/jamming

4. Confusion Matrix

	Predicted		
True	Drone Normal Jamming		
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Drone	316 13 64		
Normal	0 393 0		
Jamming	0 0 393		

5. Recommendations for Improvement

- * Collect more diverse drone data, especially from drones with unusual spectral signatures
- * Apply data augmentation specific to drone signals (time/frequency masking)
- * Fine-tune the last 20-30 layers of EfficientNet for drone-specific features
- * Consider class-weight adjustments (increase drone weight)